Data Dictionary

Ordinal-Probit Latent-Class Trajectory Model

File: ordinal_probit.sas Traj2 (released core) **Updated:** 14 October 2026

1. Overview

This file fits a single-outcome, class-specific cumulative-probit trajectory model for bounded ordinal outcomes (for example ADL, frailty, cognitive scores), by maximum likelihood in PROC NLMIXED. Including the file defines the macros and runs nothing.

Threshold parameterization. The first threshold is fixed at 0 and is *not* a parameter, so `i*1` is never emitted. The free threshold parameters are the increments `i*2` to `i*(m-1)`, added cumulatively. Each increment is held strictly positive by a `BOUNDS` statement the fitting macro generates, which is what keeps the categories ordered within a class. The increments are **not** exponentiated.

```
th1_c = 0                       (fixed, not estimated)
thj_c = th(j-1)_c + i_cj        j = 2 .. m-1
BOUNDS  i_c2 > 0, i_c3 > 0, ...  (generated per class)
```

2. Datasets

NAME	DESCRIPTION
sim_long / sim_wide	Long and wide outputs of the optional simulator (%sim_data).
BASE_FILE_SRS	Wide input table with BENE_ID, quar1..quarT holding the values 1..T, and the ordinal outcome columns.
<prefix>_EST_K<k>	Parameter estimates for a K-class fit.
<prefix>_FIT_K<k>	Fit statistics (AIC, BIC, -2 log L).
<prefix>_ESTS_K<k>	Additional estimates: thresholds on the natural scale, threshold2 to threshold(m-1). threshold1 is fixed at 0 and is not reported.
<prefix>_POST_K<k>	From %ordprob_mix_post_one . One row per subject: post1..postK, maxpost, classnum, class.
<prefix>_CLASS_K<k>	From %ordprob_mix_post_one . One row per class: n, percent, pie, app, occ.
mix_props	Estimated mixing proportions for the class-proportion plot.
pred_long	Predicted E[Y_t | class] over time for the trajectory plot.

3. Macro Arguments (%ordprob_mix_fit_one)

NAME	TYPE / DEFAULT	DESCRIPTION
data	name, BASE_FILE_SRS	Input wide-format dataset.
id	name, BENE_ID	Subject identifier.
yvars	varlist, required	Repeated-measure outcome vars (e.g. Y1_1-Y1_12).
tvars	varlist, required	Time-index vars (e.g. quar1-quar12), holding 1..T.
ttotal	integer, 12	Number of time points.
m	integer, 4	Number of ordinal categories.
ycodes	list, 0 1 2 3	Category codes.
deg	integer, 2	Polynomial degree of class trajectories.

NAME	TYPE / DEFAULT	DESCRIPTION
k	integer, 4	Number of latent classes (C).
qpoints	integer, 40	Quadrature points.
maxiter	integer, 1000	Maximum optimizer iterations.
tech	name, dbldog	PROC NLMIXED optimization technique. Double dogleg, because the increments are bounded below by zero.
start_values	text, optional	User starting values. Supply increments only, from `i*2` upward, and never `alpha0_A`. Left blank, the increments default to **1**, not 0. See the note below.
bounds	text, optional	Extra parameter bounds, combined with the generated increment bounds.
outestlib	libref, work	Library for output datasets.
prefix	name, ordprob_single	Prefix for output dataset names.

Starting values. A start of 0 for an increment sits exactly on its lower bound and must not be used. When `start_values=` is blank the macro starts every increment at 1, which places the thresholds at 0, 1, 2 and so on. Polynomial coefficients and mixing intercepts do start at 0.

4. Macro Arguments (%ordprob_mix_post_one)

Takes the same `data`, `id`, `yvars`, `tvars`, `ttotal`, `m`, `ycodes`, `deg`, `k`, `outestlib` and `prefix` as the fit, and must be given the same values. It reads `<prefix>_EST_K<k>` and recomputes each class log-likelihood on the same data with the same skipping rule, so the posteriors are consistent with the fit.

```
APP_j = mean posterior probability of class j over subjects assigned to class j
OCC_j = [APP_j / (1 - APP_j)] / [pie_j / (1 - pie_j)]
```

Nagin's guidance is APP of at least 0.7 and OCC of at least 5. Both are printed by the macro and stored in `<prefix>_CLASS_K<k>`.

5. Key Variables and Parameters

NAME	TYPE	DESCRIPTION
BENE_ID	Num	Subject identifier.
yv[ttotal]	Array	Ordinal outcome over time inside the model.
tv[ttotal]	Array	Time predictor inside the model, values 1..T.
cat	Num	Category index (1..m) matched from ycodes. Stays missing when the value is blank or unmatched, and that time point is skipped.
alpha0_[class]	Num	Class-membership logit (class A fixed to 0).
beta_[class][k]	Num	Degree-k polynomial coefficient of the latent index eta.
eta_[class]	Num	Class-specific latent index at time t.
i[class]j (j>=2)	Num	Threshold increment, held positive by BOUNDS. There is no `i[class]1`: the first threshold is fixed at 0 and is not a parameter.
th[j]_[class]	Num	Cumulative threshold j. th1 = 0, thj = th(j-1) + i[class]j.
z[j]_[class]	Num	Category-j probability from probit differences.
pie[c]	Num	Marginal class-membership probability (softmax).
llik[c]	Num	Per-class log-likelihood accumulator.
mix	Num	Mixture likelihood passed to the MODEL statement.

NAME	TYPE	DESCRIPTION
`post[c]`	Num	Posterior probability of class c per subject, from `%ordprob_mix_post_one`.
`maxpost / classnum / class`	Num / Char	Largest posterior, its index, and the class letter: the modal assignment.
`app / occ`	Num	Average posterior probability and odds of correct classification, per class.

6. Macros

NAME	PURPOSE
`%sim_data`	Optional simulator: BIN4, TPOIS4, or CAT5 ordinal DGP.
`%build_base_from_simwide`	Builds BASE_FILE_SRS with quar1..quarT = 1..T.
`%CL`	Maps a class index to its letter label (A..O).
`%_index_override_pairs`	Flags user-supplied starts so defaults skip them.
`%_make_default_parms`	Builds the default PARMS list: 0 for mixing intercepts and polynomial coefficients, 1 for threshold increments, skipping overrides.
`%ordprob_mix_fit_one`	Fits the ordinal-probit model (single PARMS statement).
`%ordprob_mix_post_one`	Posterior class membership, modal assignment, class sizes with APP and OCC.
`%ordprob_mix_plot_one`	Class-proportion barplot and predicted-trajectory curves.

7. Notes and Assumptions

- Missing time points are allowed and skipped in the likelihood (missing at random). A value present but outside `ycodes` is skipped the same way, silently, so check the coding before fitting.
- Identification: the first threshold is fixed at 0 with `beta_[class]0` free. Supply only increments `i[class]2` to `i[class](m-1)` in `start_values`, never the first threshold and never the class A mixing logit.
- Ordered categories are guaranteed by the generated `BOUNDS` on the increments, not by exponentiation.
- `cap` in `%sim_data` sets the highest category index; `cap=3` yields values {0,1,2,3}, matching `m=4` and `ycodes=0 1 2 3`.
- Fit statistics across candidate `k` values support model selection by BIC.
- Run `%ordprob_mix_post_one` after the fit to obtain APP and OCC. Neither is produced by the fit itself.